PLNKernel

Calculate one grounded proposal. Own no beliefs.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A guarded calculator: the same exact inputs enter a pinned formula, and the result goes back to the evidence owner for acceptance.

Why it matters: Separating numerical proposals from admission prevents a runtime or formula replacement from silently acquiring authority over beliefs, budgets or search.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

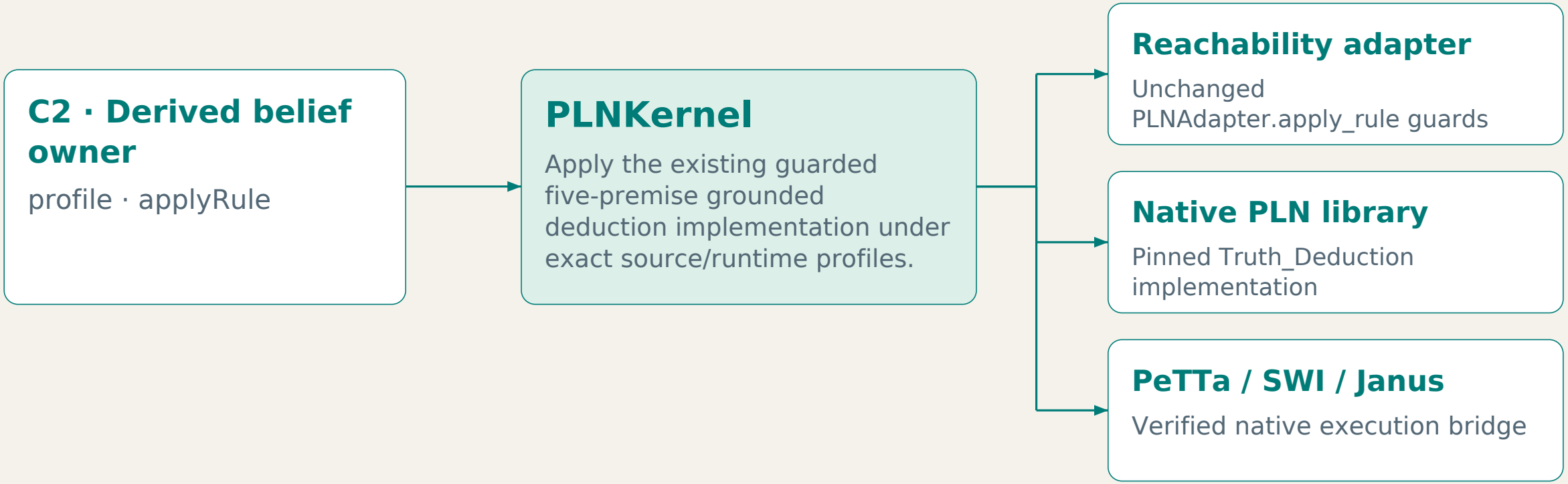
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		**PLNKernel**	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Apply the existing guarded five-premise grounded deduction implementation under exact source/runtime profiles.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

No mutable domain state

A pure proposal interface; selected runtime identity is explicit

Guarded execution boundary

Validate inputs, source/profile compatibility and returned proposal

OUTSIDE THIS RESPONSIBILITY

Admission and provenance authority

C2 validates exact claim/lineage and commits the result

Search and work budgets

No PLN.Query/Derive loop, agenda, task generation or attention mutation

grounded-five-premise-deduction/1 · fusion-petta-swi-10.1.14-deduction/1

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
profile	Identify model, rule, runtime and pinned sources	Compatibility is explicit before use
applyRule	Evaluate P, Q, R, P⇒Q, Q⇒R	One grounded proposal for P⇒R
RevisionMath.revise	Separate narrow revision dependency	Not an extra kernel method or general inference loop

One bounded grounded rule; heuristic truth model; no belief, budget or agenda state.

A worked example through the boundary

REAL native PeTTa / PLN · generated exact premises; no replacement mathematics

01 · STEP

Bind premises

Exact historical supports for five grounded roles.

02 · STEP

Validate

Check domains, bindings and probability consistency.

03 · STEP

Execute native

Run unchanged guarded Truth_Deduction.

04 · STEP

Return proposal

C2 rechecks authority and decides admission.

What the example establishes

Reusing mathematics requires preserving its assumptions and boundary.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Pinned mathematics

No alternative formula is substituted to obtain a desired result.

I2

Exact premise vector

Historical support identities are not replaced by latest beliefs.

I3

Refuse invalid domains

Nonfinite or inconsistent inputs fail guarded checks.

I4

No owner writes

A successful formula result alone changes no belief or attention state.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Native process unavailable	Return explicit unavailable/failed attempt semantics.	No fixture fallback for real gate
Crash around formula return	C2/C4 journals govern recovery and spent attempts.	Kernel is not the recovery owner
Source/runtime mismatch	Reject incompatible selection before admission.	Historical SWI 10.0.1 is unrun

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

5 → 1

Premises → grounded proposal

One rule, not general inference

0.68 / 0.3136

Recorded deduction strength / confidence

Generated golden; tolerance 1e−12

0

Mutable domain owners in kernel

Architectural invariant, not performance

The numerical golden checks implementation agreement, not probabilistic calibration or theorem validity. No isolated native-call latency or large-rule benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: separate process startup from warm native calls; fix exact valid vectors and refusal cases; report p50/p95, guard cost and timeout behavior without changing the formula.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Pinned Reachability, PLN, PeTTa, SWI 10.1.14 and Janus execute the native formula.
FIXTURE / LOCAL TEST POLICY	Generated premise vectors and independent expected numerical goldens.
BLOCKED / UNSUPPORTED	No required supported gate blocked; historical SWI 10.0.1 compatibility remains unrun.
UNRUN	General rule search, calibration, broad rule coverage and cross-runtime numerical equivalence.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure the runtime boundary	Warm/cold fixed-vector experiments	Separate startup, guards and native evaluation
Qualify a new runtime pin	Golden, refusal and source compatibility matrix	No silent numerical or exception changes
Add another rule deliberately	Explicit rule profile and lineage/admission review	Each new rule has independent goldens and refusal tests

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Reusing mathematics requires preserving its assumptions and boundary.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3.py <env> --output <fresh>
```

Build pinned providers with build_m3.py first; use a fresh source checkout and explicit runtime selection. The wrapper includes both guard and actual-native suites.

`fusion/adapters/pln_deduction.py`
`integration/pln/test_deduction.py`
`integration/pln/deduction-sources.json`
`docs/module-interfaces/m3-evidence.md`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.